

Goals for this section:

- Demonstrate how to create and configure Group Policy Objects (GPOs)
- Demonstrate how to link GPOs to OUs
- Considering this lab is in a school setting, I will configure GPOs for the entire domain, specific OUs, and specific groups
- Note: I will be using the Administrator account on the Domain Controller for simplicity and demonstration purposes. In the real world, it would be better to delegate limited administrative privileges to a user account for security purposes

Creating Group Policy Objects (GPOs)

General Navigation

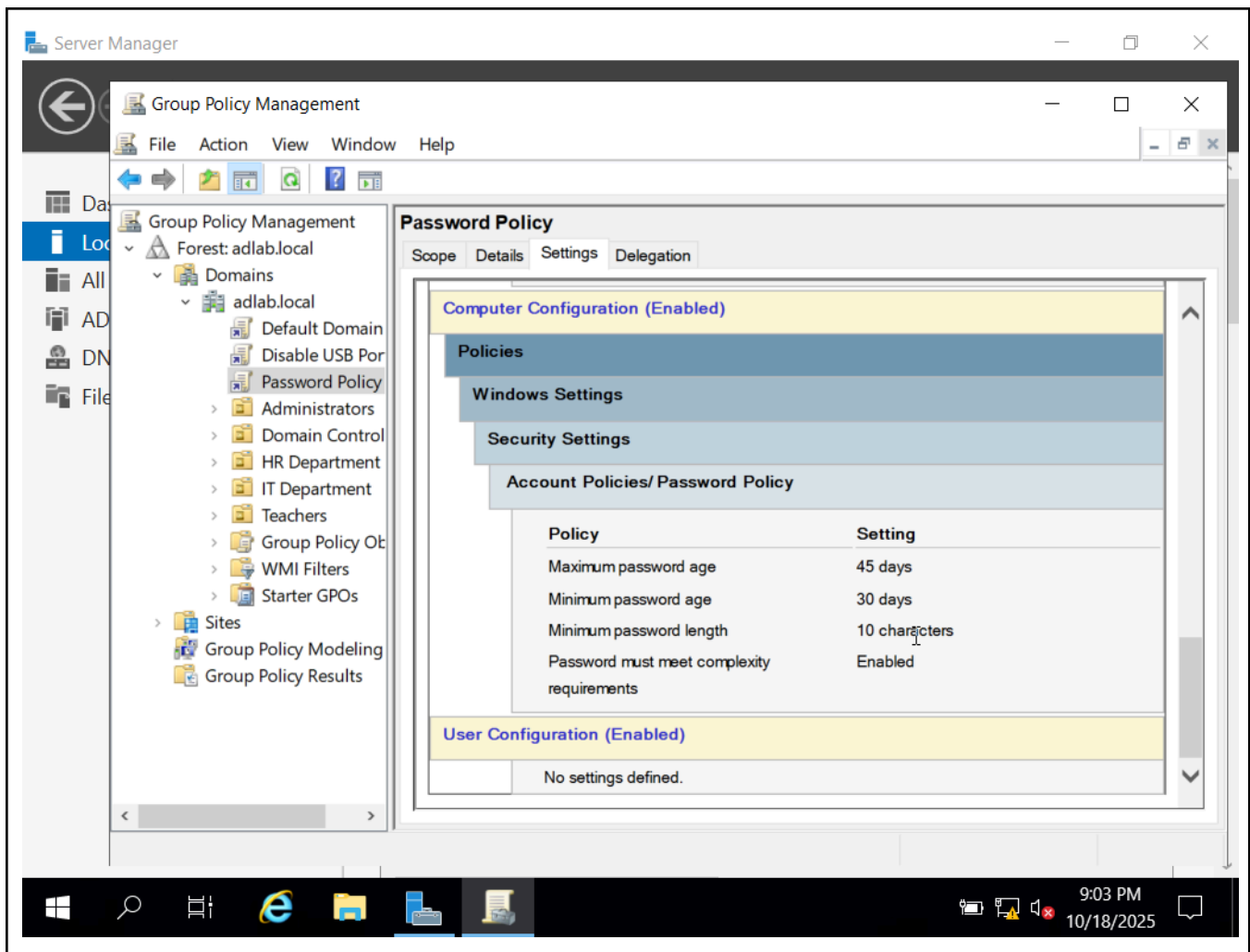
- Open "Group Policy Management"
- Click the ">" arrow next to "Forest:..." → In the dropdown menu, click the ">" next to "Domains" → In the dropdown menu, click the ">" next to the server (adlab.local)

Creating a domain-wide GPO

- Right click the domain → "Create a GPO in this domain, and Link it here..." → Name the GPO
- Right click the newly created GPO → Edit → The next step varies, but you will have to find the type of policy you need for the GPO, and configure it to the appropriate policies and permissions
- I will create **2** domain-wide GPOs applying to all users for this lab:

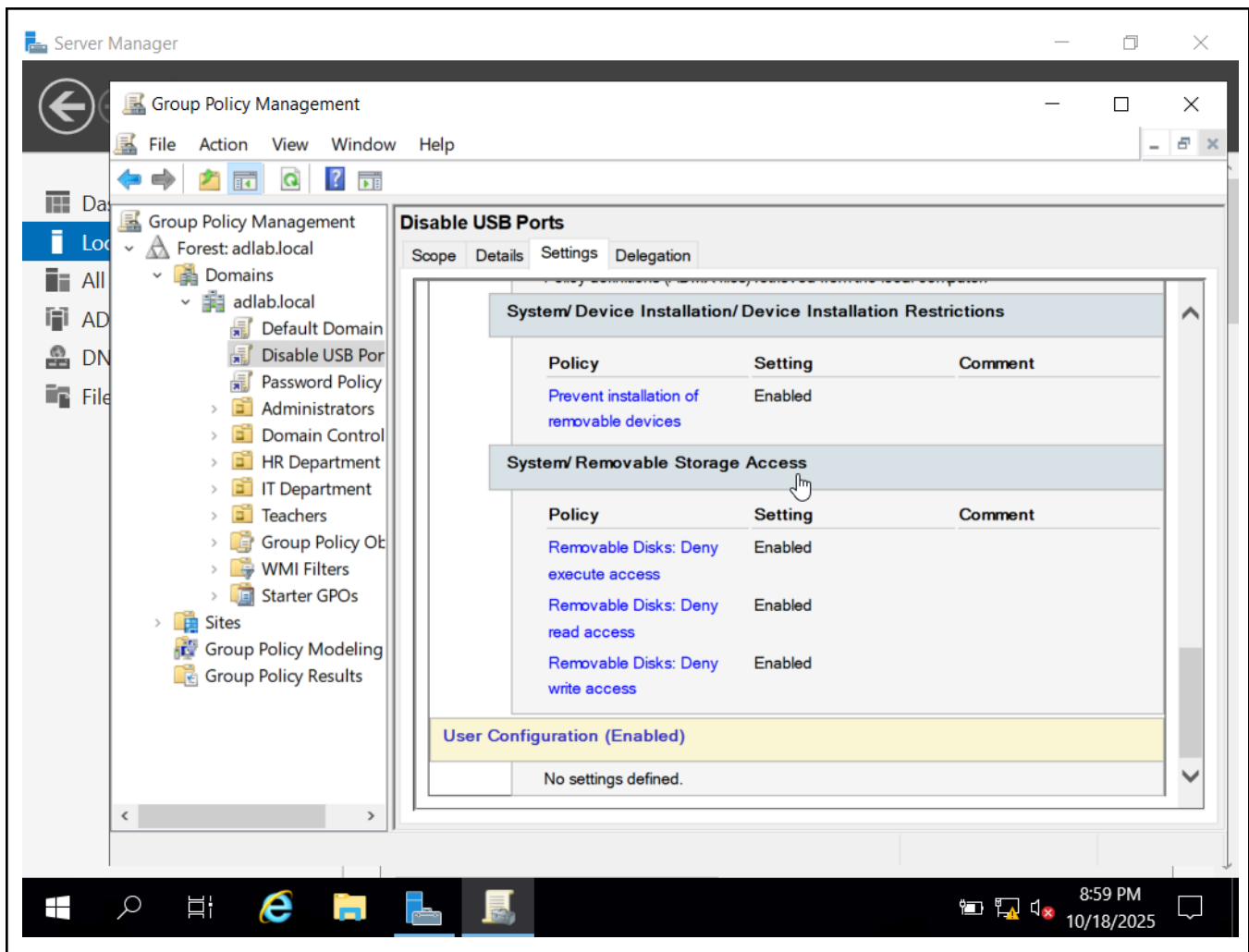
1. Password Policy

- a) Maximum age: 45 days
- b) Minimum password length: 10 characters
- c) Password must meet complexity requirements: Enabled



2. Disable USB Ports

- Prevent installation of removable devices: Enabled
- Removable Disks - Deny execute access: Enabled
- Removable Disks - Deny read access: Enabled
- Removable Disks - Deny write access: Enabled

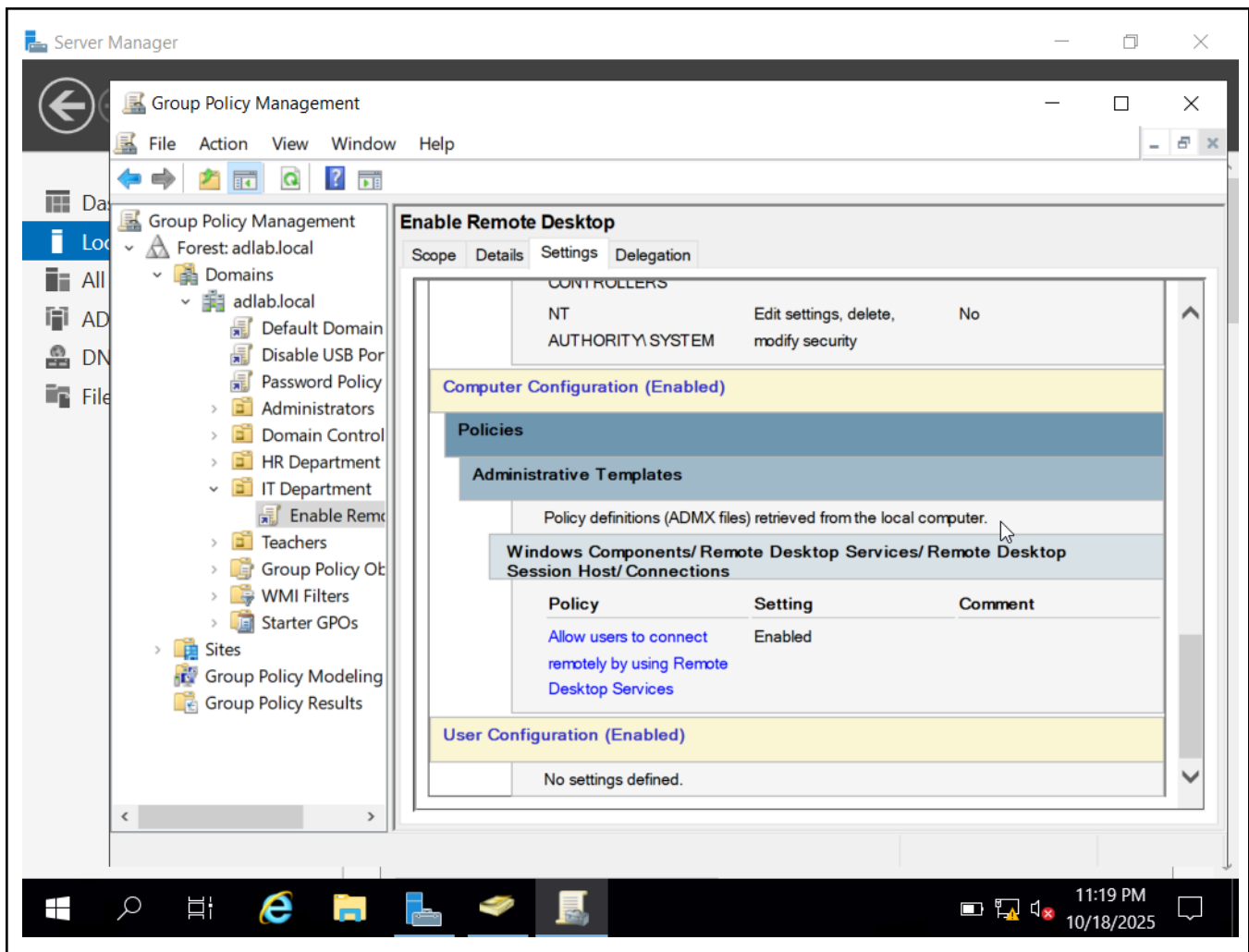


Creating a GPO for specific OUs

- Right click the OU → “Create a GPO in this domain, and Link it here...” → Name the GPO
- Right click the newly created GPO → Edit → The next step varies, but you will have to find the type of policy you need for the GPO, and configure it to the appropriate policies and permissions and to limit the scope to the OU
- I will create **4** GPOs, one for each group in the domain for this lab:

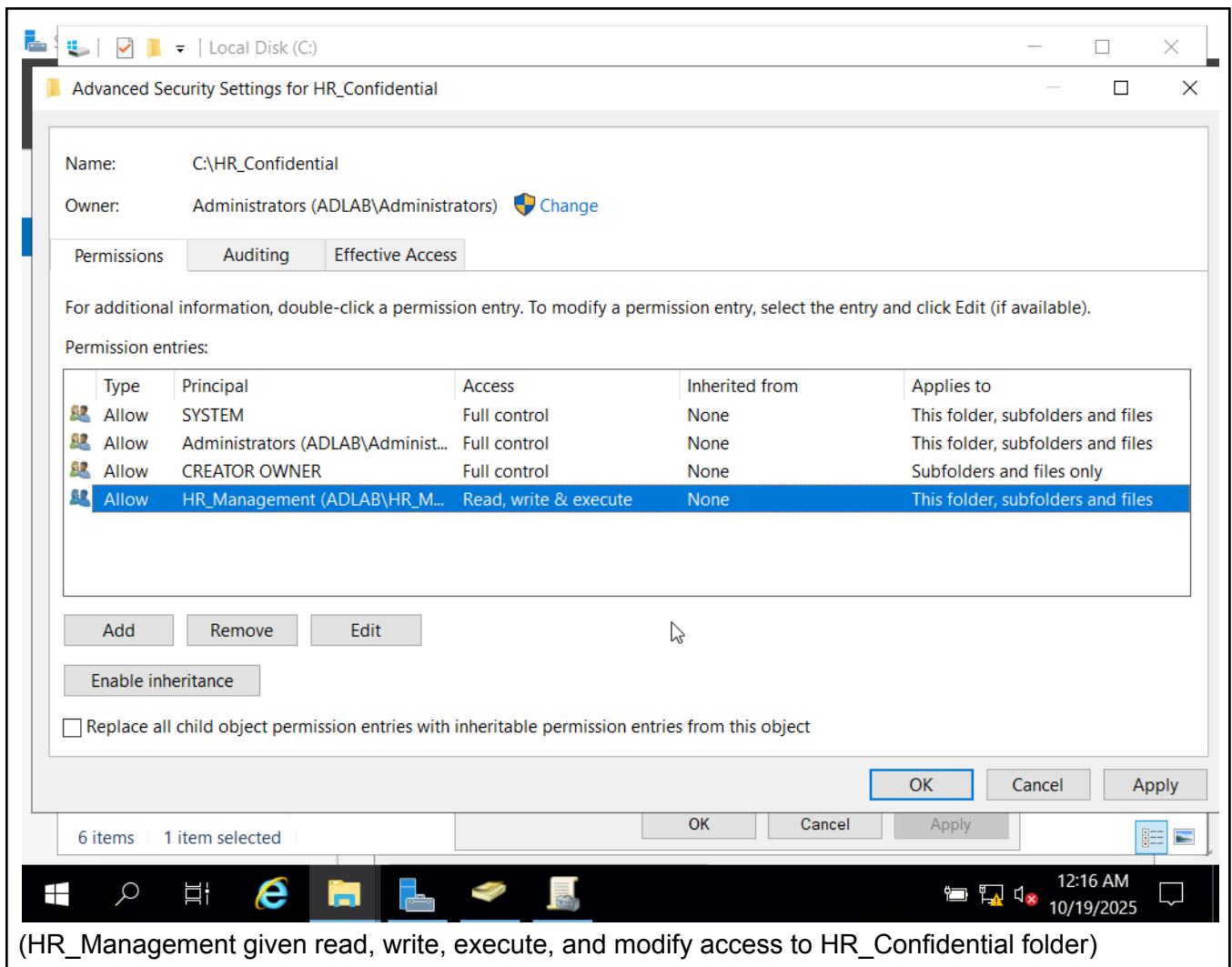
1. Group: IT Admins | Policy: Enable Remote Desktop

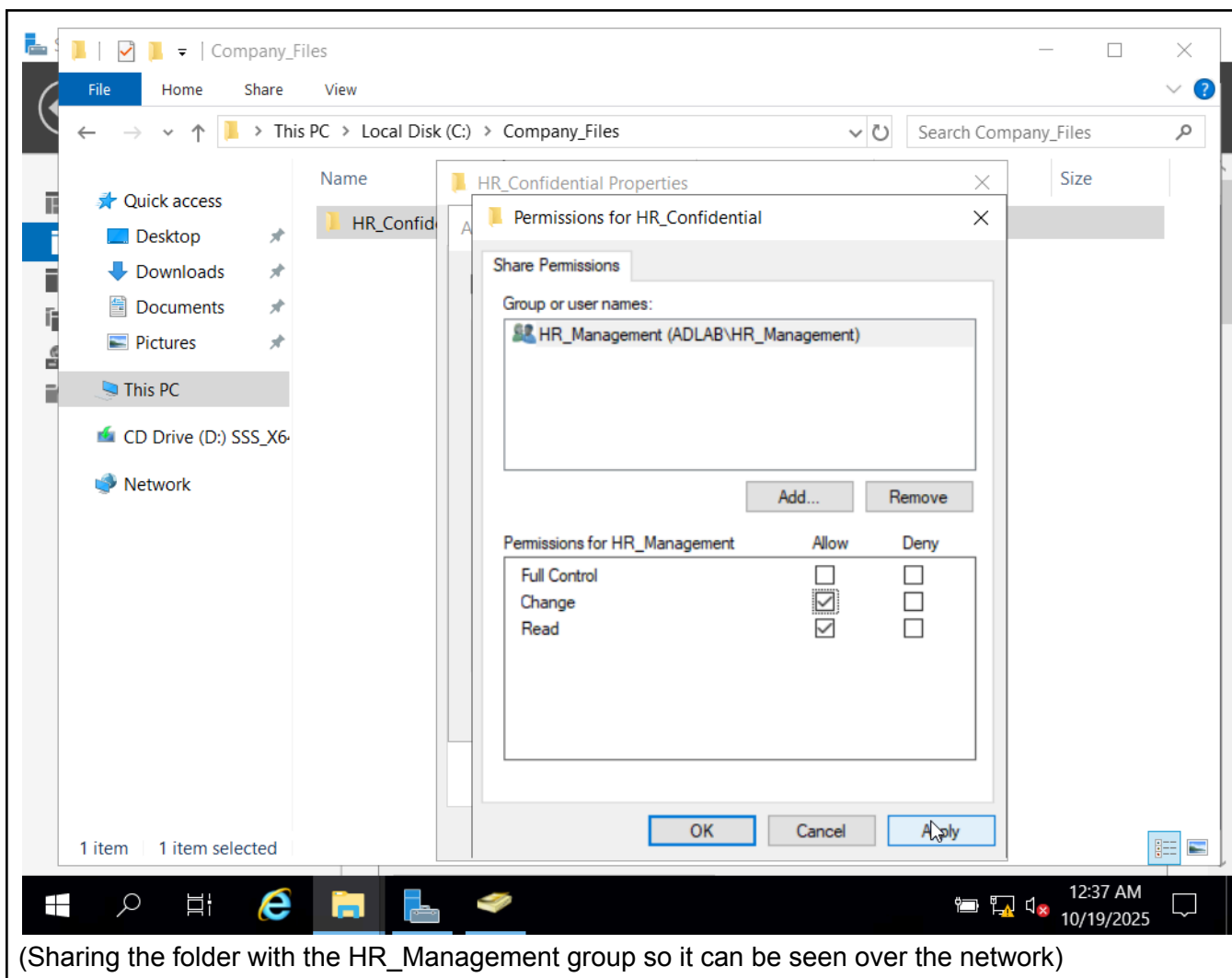
- Computer Configuration → Policies → Administrative Templates → Windows Components → Remote Desktop Services → Remote Desktop Session Host → Connections
- Allow users to connect remotely by using Remote Desktop Services: Enabled
- Select the “scope” ribbon → In security filtering, Add IT Admins and remove Authenticated Users

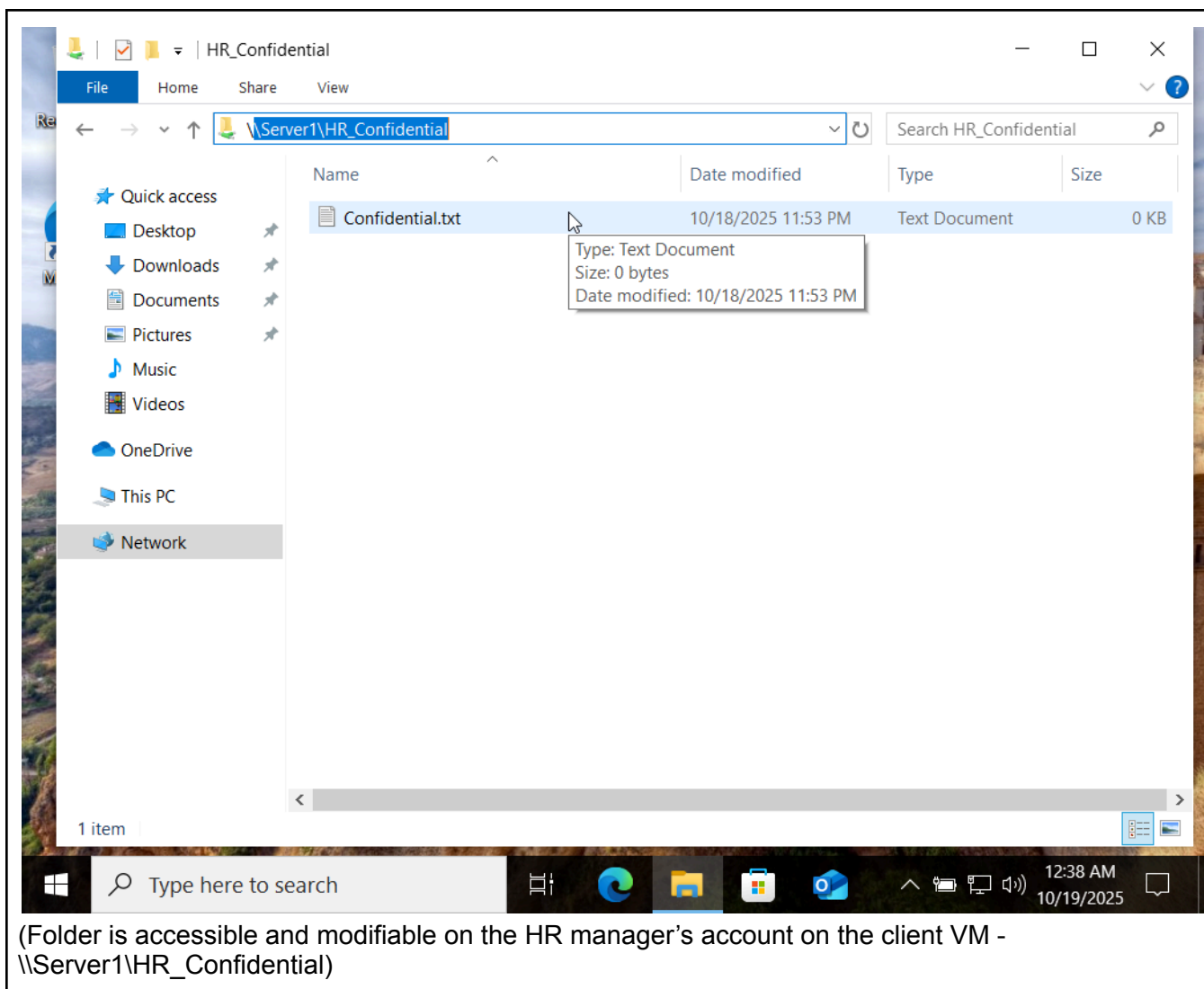


2. Group: HR Management | Policy: Map a confidential folder

- Create a folder on the server: (Path: C:\Company_Files\HR_Confidential)
- Right click folder → Properties → Security ribbon → Advanced
- If "Users" have any permissions, Remove "Users" from access: Disable inheritance → Remove "Users" permissions
- Add → Security group (HR_Management)
- Allow permissions:
 - Modify
 - Read & Execute
 - List folder contents
 - Read
 - Write
- Right click folder → Properties → Sharing ribbon → Advanced Sharing... → Select "Share this folder" → Permissions → Remove "Everyone" → Add security group (HR_Management)



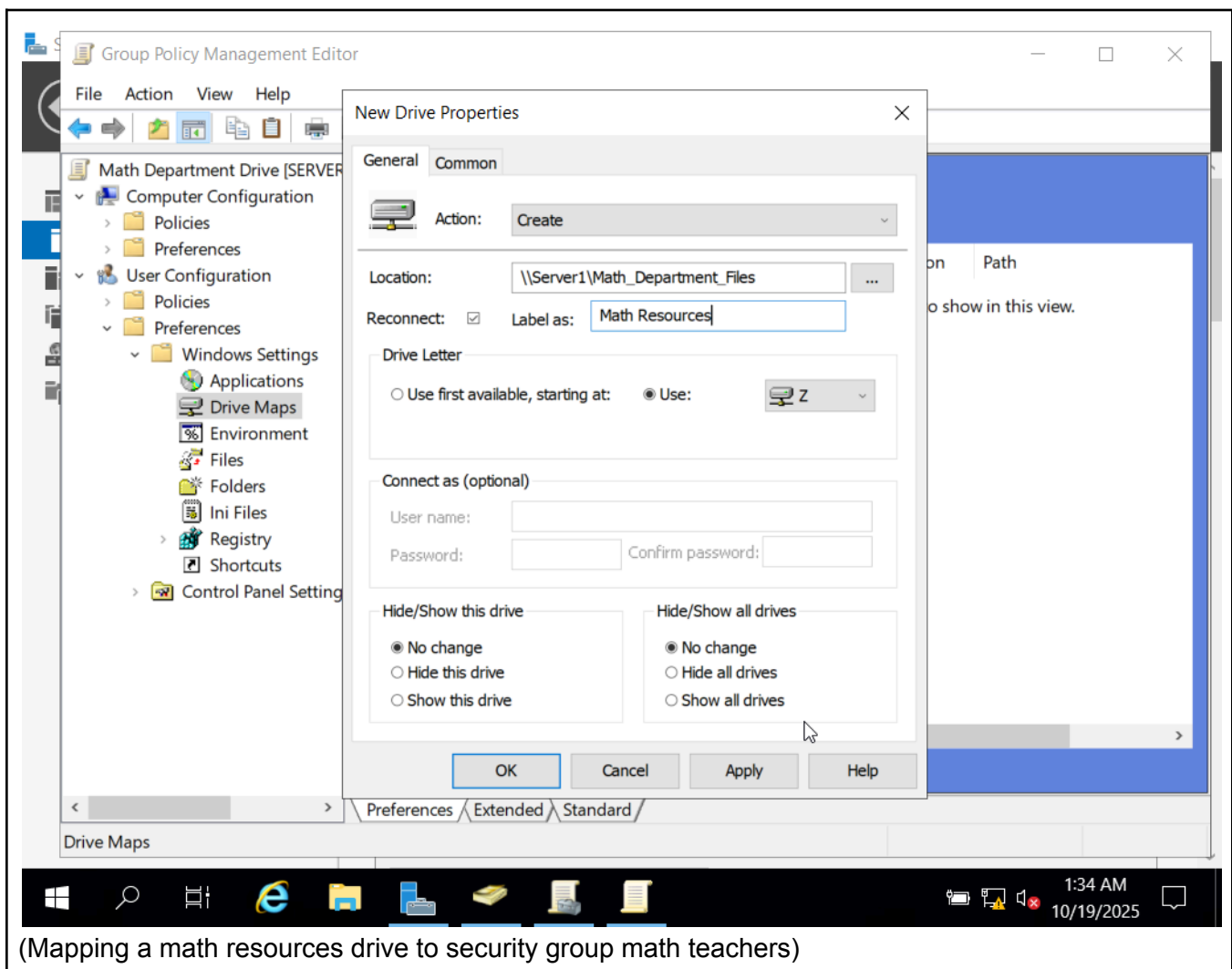


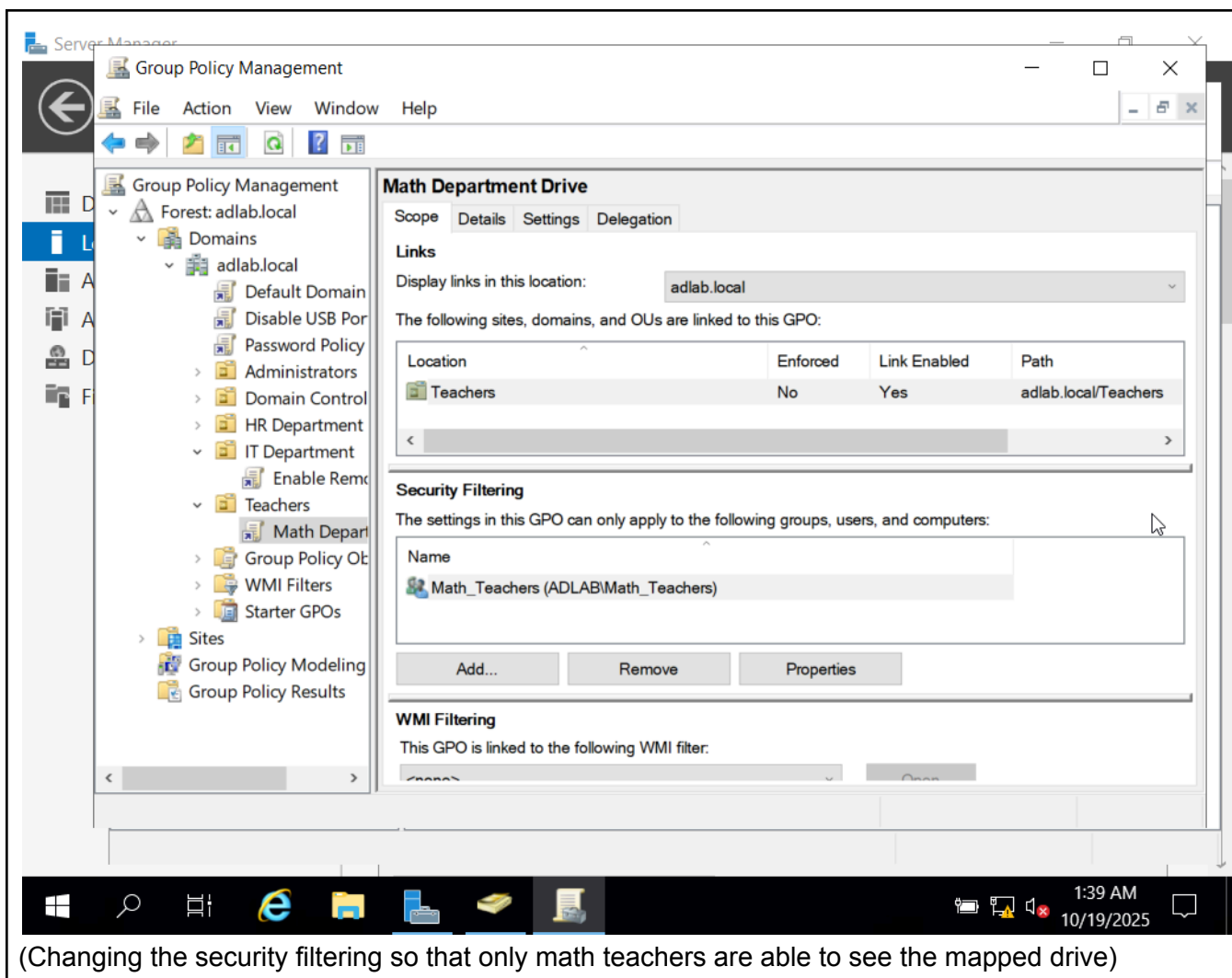


3. Group: Math Teachers | Policy: Map a network drive

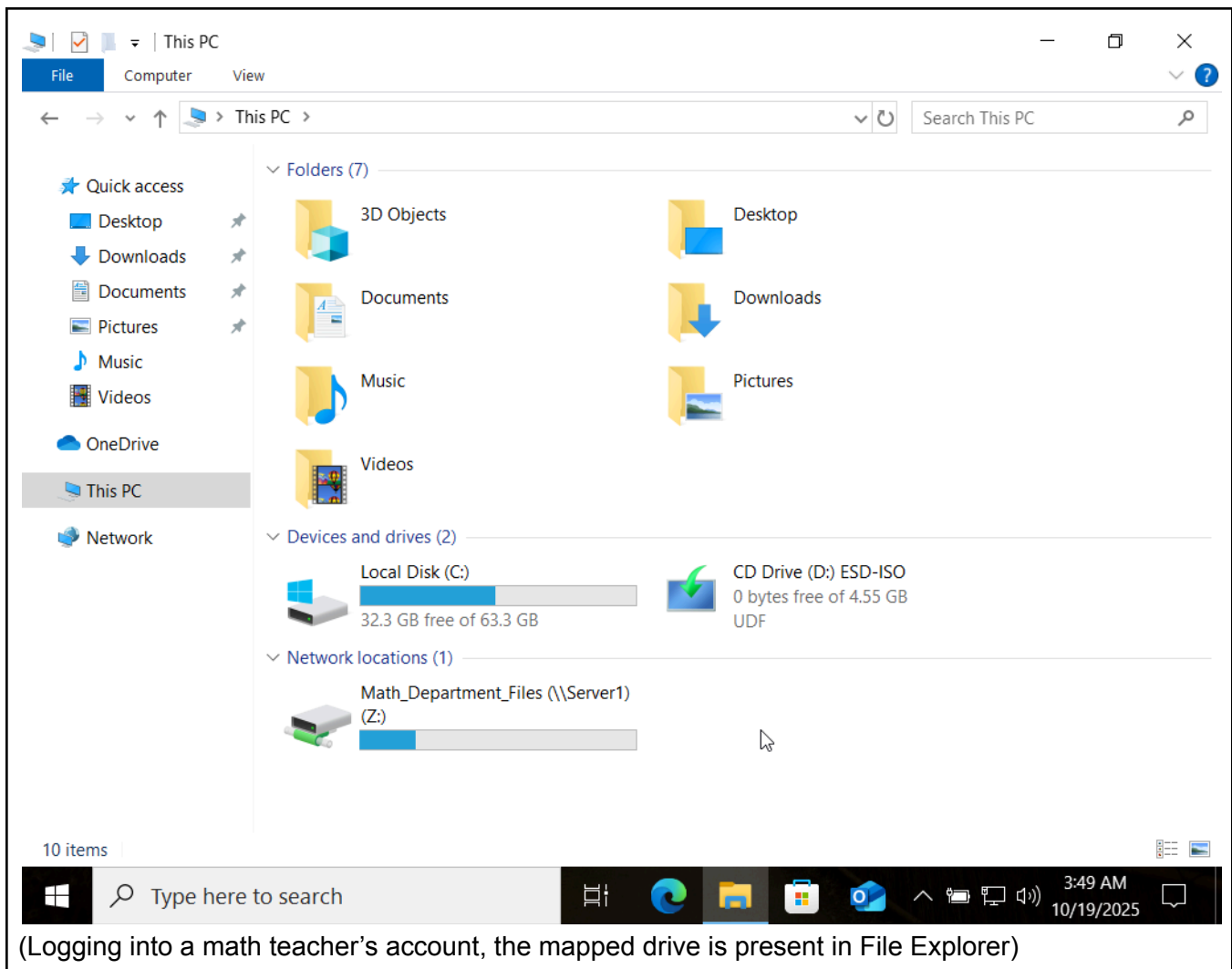
- Create a drive on the server: (Path: C:\Company_Files\Math_Department_Files)
- Right click folder → Properties → Security ribbon → Advanced
- If "Users" have any permissions, Remove "Users" from access: Disable inheritance → Remove "Users" permissions
- Add → Security group (Math_Teachers)
- Allow permissions:
 - Modify
 - Read & Execute
 - List folder contents
 - Read
 - Write
- Right click folder → Properties → Sharing ribbon → Advanced Sharing... → Select "Share this folder" → Permissions → Remove "Everyone" → Add security group (Math_Teachers)

- g) Create a GPO to map the drive: Right click Teachers OU → Create a GPO and Link... → Name it (Math Department Drive)
- h) Right click new GPO → Edit → User Configuration → Preferences → Windows Settings → Drive Maps
- i) Right click Drive Maps → New → Mapped Drive
- j) Initialize with the properties:
 - **Action:** Create
 - **Location:** \\Server1\Math_Department_Files
 - **Reconnect:** Checked
 - **Use Drive Letter:** Z
 - **Label as:** Math Resources
- k) Scope the file to the appropriate security group (Math teachers): Expand the OU (Teachers) → Click the GPO (Math Department Drive) → Scope ribbon → In Security Filtering, Remove “Authenticated Users” → Add the appropriate Group (Math_Teachers)





(Changing the security filtering so that only math teachers are able to see the mapped drive)



4. Group: Deans | Policy: Audit and log logon activity and file access for Deans group

- a) Open GPM → Select OU (Administrators) →: Create new GPO and Link... → Name it (Deans Audit Policy)
- b) Right click new GPO → Edit → Computer Configurations → Policies → Windows Settings → Security Settings → Advanced Audit Apolicy Configuration → Audit Policies
- c) Category:
 - **Account Logon**
 - **Audit Credential Validation:** Success and Failure (logs successful and failed authentication attempts)
 - **Logon/Logoff**
 - **Audit Logoff:** Success (logs successful sign out attempts)
 - **Audit Logon:** Success and Failure (logs successful or failed sign in attempts)
 - **Object Access**

→ **Audit File System:** Success and Failure (logs folder access, modification, or deletion)

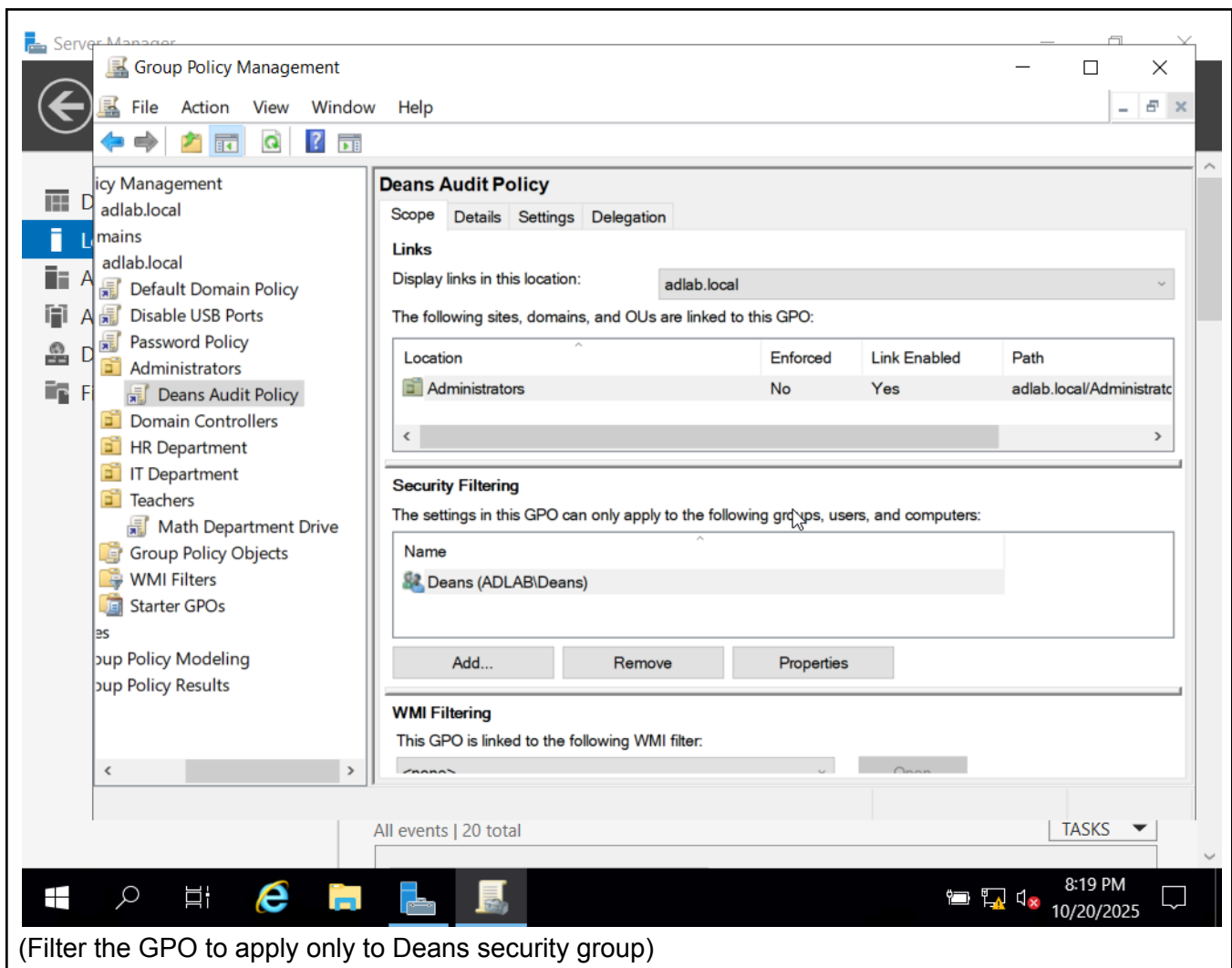
- **Policy Change**

→ **Audit Authentication Policy Change:** Success and Failure (logs all changes and modifications to domain)

→ **Audit Authorization Policy Change:** Success (logs changes of user rights and security policy changes)

d) Logon to a Dean account and run some actions

e) Back on the Domain Controller, search for “Event Viewer” → Windows Logs → Security



Group Policy Management

File Action View Window Help

Policy Management

- adlab.local
- adlab.local
- Default Domain Policy
- Disable USB Ports
- Password Policy
- Administrators
- Deans Audit Policy**
- Domain Controllers
- HR Department
- IT Department
- Teachers
- Math Department Drive
- Group Policy Objects
- WMI Filters
- Starter GPOs

Deans Audit Policy

Scope Details Settings Delegation

Advanced Audit Configuration

Account Logon

Policy	Setting
Audit Credential Validation	Success, Failure

Logon/ Logoff

Policy	Setting
Audit Logoff	Success
Audit Logon	Success, Failure

Object Access

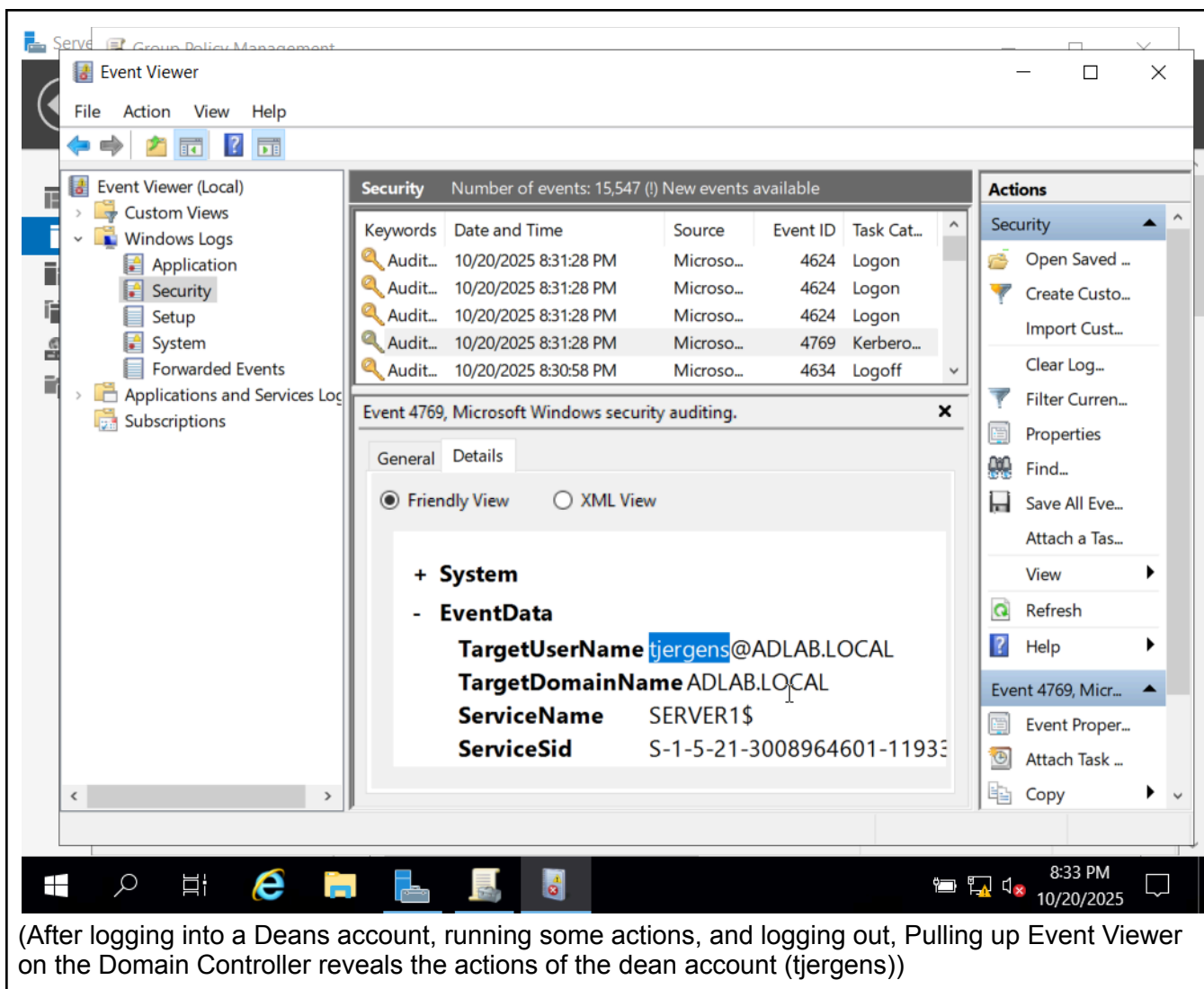
Policy	Setting
Audit File System	Success, Failure

Policy Change

Policy	Setting
Audit Authentication Policy Change	Success, Failure
Audit Authorization Policy Change	Success

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(Deans Audit Policy GPO configuration policies)



(After logging into a Deans account, running some actions, and logging out, Pulling up Event Viewer on the Domain Controller reveals the actions of the dean account (tjergens))